Homework 6

MEGN 544/ROBO 554 Robot Mechanics: Kinematics, Dynamics, and Control

Dr. George Kontoudis
Department of Mechanical Engineering
Colorado School of Mines

Fall 2026

Problem 1. Let a manipulator with one prismatic and five rotational joints to be characterized by the following DH table.

Link	a	d	α	θ
1	a_1	0	0	$\star$
2	a_2	0	0	$\star$
3	0	$\star$	0	0
4	0	0	$\pi/2$	$\star$
5	0	0	$-\pi/2$	$\star$
6	0	0	0	$\star$

The forward kinematics is given by,

$$
{}^0\boldsymbol{T}_6 = \begin{bmatrix} c_{124}c_5c_6 - s_{124}s_6 & -s_{124}c_6 - c_{124}c_5s_6 & -c_{124}s_5 & c_{12}a_2 + c_1a_1 \\ s_{124}c_5c_6 + c_{124}s_6 & c_{124}c_6 - s_{124}c_5s_6 & -s_{124}s_5 & s_{12}a_2 + s_1a_1 \\ c_6s_5 & -s_5s_6 & c_5 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix},
$$

where $c_{124} = \cos(\theta_1 + \theta_2 + \theta_4)$. Given the homogeneous transformation,

$$
{}^0\boldsymbol{T}_6 = \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix},
$$

the inverse kinematics solution becomes,

$$
\begin{aligned}
\theta_2 &= \pm 2\text{atan}\left(\sqrt{\frac{(a_1 + a_2)^2 - (p_x^2 + p_y^2)}{(p_x^2 + p_y^2) - (a_1 - a_2)^2}}\right) \\
\theta_1 &= \text{atan2}(p_y, p_x) - \text{atan2}(a_2 \sin\theta_2, a_1 + a_2\cos\theta_2) \\
d_3 &= p_z
\end{aligned}
$$

$$\theta_5 = \text{atan2}\left(\pm\sqrt{r_{13}^2 + r_{23}^2}, r_{33}\right)$$
$$\theta_4 = \text{atan2}\left(-\frac{r_{23}}{\sin\theta_5}, -\frac{r_{13}}{\sin\theta_5}\right) - \theta_1 - \theta_2$$
$$\theta_6 = \text{atan2}\left(-\frac{r_{32}}{\sin\theta_5}, \frac{r_{31}}{\sin\theta_5}\right)$$

Let the link lengths $a_1 = 1$ and $a_2 = 0.5$, the initial manipulator pose at $t = 0$ seconds,

$${}^0\boldsymbol{T}_6(0) = \begin{bmatrix} 1 & 0 & 0 & 1.00 \\ 0 & 0 & -1 & 0.25 \\ 0 & 1 & 0 & 0.25 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

and the final manipulator pose at $t = 2$ seconds,

$${}^0\boldsymbol{T}_6(2) = \begin{bmatrix} 0 & -1 & 0 & 0.50 \\ 1 & 0 & 0 & -0.50 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

(a) Write a MATLAB function to generate the coefficients for cubic interpolation between two points given zero initial and final velocity. Comment in the script as in the project. and provide a print-out of the code.

(b) Generate a MATLAB function to interpolate between two vectors, given cubic interpolation coefficients, an initial time, and a list of times for evaluation. Comment the script as in the project and provide a print-out of the code.

(c) For the following parameterizations, generate a figure that shows the path in joint variable space from the initial configuration to the final configuration and a figure that shows the Cartesian path from the initial configuration to the final configuration. Provide a descriptive title, label all axis, and use different line types (with a legend) for plots with more than one trace. Also write a script using the provided `drawArm.m` file to play the motion in a movie. Discuss the differences and similarities. *Hint: Use pause function within a for loop.*

i. In parameter space (θ,d) from initial joint values to final.

ii. In Cartesian/angle-axis space.

iii. In Cartesian/quaternion space (make sure you normalize the interpolated quaternion at each sub point).

iv. In twist space.